Prepared By ASCOL CSIT 2070 Batch

Tribhuvan University

Institute of Science and Technology

2070

✡

Bachelor Level/ Second Year/ Third Semester/ Science — Full Marks: 60

Computer Science and Information Technology (CSc. 203) — Pass Marks: 24

(Operating System) — Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Section A

Attempt any two questions: (2x10=20)

1. For the processes listed in following table, draw a Gantt chart illustrating their execution using:
   a) First-Come-First-Serve
   b) Shortest-Job-First
   c) Shortest-Remaining-Time-Next
   d) Round-Robin (quantum = 2)
   e) Round-Robin (quantum= 1)

Processes	Arrival Time	CPU Time
A	0.000	3
B	1.001	6
C	4.001	4
D	6.002	2

What is the turnaround time for each algorithm?

OR

What do you mean by disk management? What the major difference between error handling and formatting.

2. How many page faults occur for each of the following page replacement algorithms for the reference string 0 1 7 2 3 2 7 1 0 3 with four page frames and eight pages. Suppose all frames are initially empty.
   a) Optimal replacement
   b) FIFO replacement
   c) LRU replacement
   d) Clock replacement
3. Suppose that the disk drive has 50 cylinders, numbered from 0 to 49. The drive currently serving the request at cylinder 20 and the previous request was at cylinder 25. The queue of pending request is 10, 22, 20, 2, 40, 6 and 38 in the order. A seek takes 6msec per cylinder moved. How much seek time is needed for the following disk-scheduling algorithms?
   a) First-Come, First-Served
   b) Shortest Seek Time First
   c) SCAN
   d) LOOK

Section B

Attempt any eight questions: (8x5=40)

4. Define the essential properties of following types of operating systems.
   a) Batch
   b) Interactive
   c) Time sharing
   d) Real time
   e) Handheld
5. Describe how multithreading improve performance over a singled-threaded solution.
6. "Using Semaphore is very critical for programmer". Do you support this statement? If yes. Prove the statement with some fact. If not. Put your view with some logical facts against the statement.
7. Students working at individual PCs in a computer laboratory send their files to be printed by a server which spools the files on its hard disk. Under what conditions may a deadlock occur if the disk space for print spool is limited? How many the deadlock be avoided?
8. What are Segmentation and Paging? Why they are sometimes combine into one scheme?
9. What are the differences between the trap and interrupt? What is the use of each function?
10. What is "device independence"? Define.
11. Explain how file allocation table (FAT) manage the files. Mention the merits and demerits of using FAT.
12. Write short notes on (any two):
    a) System programs
    b) Race condition
    c) Windows file system